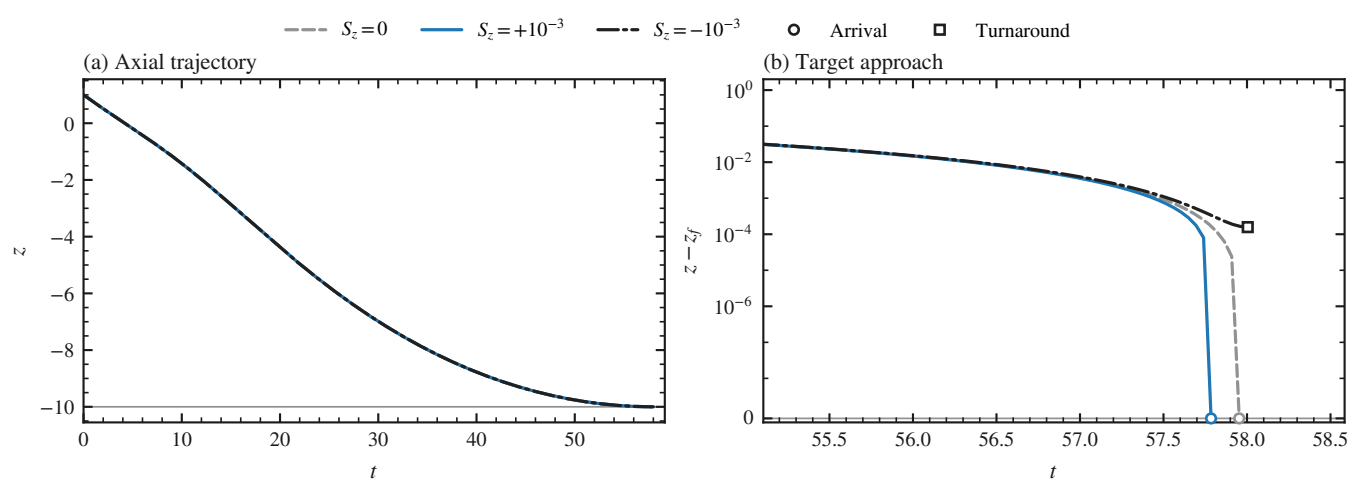
$S_z = 0$
$S_z = +10^{-3}$
$S_z = -10^{-3}$
Arrival
Turnaround
(a) Axial trajectory
$z$
$t$
(b) Target approach
$z - z_f$
$t$